Investigation for Auricular Syphilis

Any patient presenting with unexplained sudden hearing loss should be screened for syphilis.

Management

Individuals diagnosed with syphilis are at increased risk of acquiring other sexually transmitted infections (STIs). All patients with syphilis should be tested for HIV and hepatitis C virus (HCV), particularly if local epidemiology or individual risk factors support such testing. A comprehensive STI assessment is recommended for all cases.

Evaluation and vaccination for hepatitis B should also be considered as appropriate.

General Remarks

- A treponemicidal concentration of antimicrobial must be achieved in the serum; in cases of neurosyphilis, this includes the cerebrospinal fluid (CSF). A penicillin level of >0.018 mg/L is considered treponemicidal, though this is significantly lower than the maximally effective in vitro concentration of 0.36 mg/L.

- The duration of treponemicidal antimicrobial levels should last at least 7–10 days to cover multiple division cycles of *Treponema pallidum* (approximately 30–33 hours). Longer treatment durations are required as the duration of infection increases, due to higher relapse rates observed with shorter regimens in late syphilis—likely because of slower replication of treponemes in later stages. Persistent treponemes have been documented even after apparently successful treatment, though the clinical significance of this finding remains uncertain.

- Long-acting benzathine penicillin G (BPG) at 2.4 million units intramuscularly is the first-line treatment, maintaining treponemicidal serum levels for 21–28 days. For daily parenteral therapy with procaine penicillin, a safety margin is achieved by administering treatment for 10–14 days in early syphilis and 10–21 days in late syphilis. However, high-quality clinical data on optimal dosing, duration, and long-term efficacy—especially for non-penicillin agents—are limited, even for penicillin.

- Current treatment guidelines are largely based on laboratory data, biological plausibility, practical considerations, expert opinion, case reports, and historical clinical experience.

- Parenteral penicillin is preferred over oral regimens because it ensures supervised administration and reliable bioavailability. However, oral amoxicillin combined with probenecid has demonstrated efficacy and can achieve treponemicidal concentrations in the CSF.

- Non-penicillin alternatives have been studied: - **Tetracyclines**: Doxycycline is the preferred agent due to good CSF penetration. - **Erythromycin**: Less effective and poor at crossing the blood-brain and placental barriers. - **Ceftriaxone**: Administered intramuscularly or intravenously, it has excellent CSF penetration. However, its use requires multiple doses, lacks standardized regimens, and offers no clear advantage over single-dose BPG. Daily intravenous or subcutaneous ceftriaxone may be considered in patients with bleeding disorders. - In penicillin-allergic patients, ceftriaxone may be an option, though cross-reactivity risk exists. A history of anaphylaxis to penicillin is an absolute contraindication.

- **Azithromycin**: Demonstrated treponemicidal activity in animal models and some human studies, particularly in African populations. However, resistance develops rapidly, and clinical treatment failures have been widely reported.

- Host immune response plays a critical role in disease progression: approximately 60% of untreated individuals do not progress beyond primary lesions. CSF involvement is common in early syphilis. Despite the fact that standard regimens of benzathine and procaine penicillin do not reliably achieve treponemicidal levels in the CSF, the incidence of late complications—including neurosyphilis—remains low. This suggests that host immunity contributes significantly to disease control in early syphilis.